MD ASHIKUR RAHMAN

Lead AI Engineer | Multimodal AI, Generative AI & Applied Computer Vision

Dhaka, Bangladesh | Open to Relocation and Remote Opportunities

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Summary

Lead AI Engineer specializing in multimodal AI, vision-language systems, generative AI, and applied computer vision. Currently lead a multidisciplinary team of 15+ engineers and researchers and deliver enterprise AI products used by 200+ brands. Architected computer-vision systems that have processed more than 4.5 million images globally.

Experience

Lead AI Engineer, The KOW Company — Jan 2023-Present

• Lead a multidisciplinary team of 15+ ML engineers, software engineers, and junior researchers, driving technical strategy, system architecture, engineering execution, quality standards, and cross-functional delivery.
• Lead the design, development, and deployment of scalable AI systems, translating product requirements into robust machine learning solutions.
• Direct applied research on AI hallucination, prompt safety, visual grounding, and video understanding, operationalizing research into production-ready models and evaluation frameworks.
• Develop and deploy deep learning–based computer vision and image-processing systems; publish reproducible code, datasets, and model checkpoints on GitHub and Hugging Face.

Senior Machine Learning Engineer, The KOW Company — Jul 2021-Dec 2022

• Improved object detection and segmentation performance by 20-35% across internal evaluation benchmarks; the resulting models were later deployed in Retouched.ai.
• Led 6+ client ML engagements from requirements gathering through production delivery; built offline evaluation pipelines and A/B testing workflows to validate model quality, inference performance, and business and operational outcomes.

Machine Learning Engineer, The KOW Company — Jul 2020-Jun 2021

• Built deep learning models for object recognition, image segmentation, and background-removal workflows; developed scalable preprocessing, training, and A/B testing pipelines.

Senior Software Engineer, Smart Technologies (BD) Ltd — Sep 2016-Dec 2019

• Led .NET and SQL Server supply-chain systems on a 4 TB database, reducing report generation from 20 minutes to 40-54 seconds.
• Achieved 70-75% process automation and 99.9% synchronization success for offline-capable enterprise workflows.

Software Engineer, Proggasoft — Mar 2015-Aug 2016

• Developed ASP.NET MVC features, backend services, and database integrations for DevSkill.com.

Key Projects

Retouched.ai - Object Detection and Segmentation — Production

• Developed salient-object segmentation for background removal; improved segmentation quality by 17% on internal benchmarks, reduced processing time by 30%, enabled uploads of up to 257 MB, and achieved a 2.27-second average processing time across standard workloads.
• Scaled Retouched.ai to process 4.5M+ images globally for hundreds of customers using PyTorch, U²-Net-inspired salient-object segmentation, FastAPI, and Google Cloud Platform.

Omnimage.ai - AI Image and Video Generation — Production

• Co-designed and launched image- and video-generation APIs used by 200+ brands for creative and product-image workflows.
• Engineered workflows for reference-image conditioning, asynchronous processing, prompt classification, intent routing, and automated model selection.

The Fitting Room - Cross-Brand Virtual Try-On Platform (In development)

• Conceived and co-led a unified cross-brand virtual try-on platform supporting products from 170+ brands.
• Architected a Dockerized FastAPI/Nginx backend using SQL Server, Google Cloud Storage, Redis, recommendation services, and 2D virtual try-on pipelines.

Enterprise AI Catalog Audit and Image Quality Assurance Platform (In development)

• Leading the development of an AI catalog-audit and image quality assurance platform for a major US retail client.
• Automating image QA, catalog validation, metadata checks, and content-health monitoring across DAM and CMS workflows using NLP, PyTorch, and FastAPI.

CogniX - Proprietary Multimodal Content Creation Platform (Active R&D)

• Leading early-stage R&D for multimodal product-visualization and editing workflows.
• Researching and prototyping multi-reference fusion and editing workflows for garment replacement, scene composition, product visualization, and style transfer using PyTorch, Diffusers, ComfyUI, and parameter-efficient fine-tuning (LoRA, QLoRA).

Peer-Reviewed Publications

[P1] Abdullah Ibne Hanif Arean, Niamul Hassan Samin, Md Arifur Rahman, Renu Akter Sweety, Juena Ahmed Noshin, **Md Ashikur Rahman***. “Stroke-Level Connectivity Verification: Grounding Vision-Language Models Against Topology Hallucination in Diagram Understanding.” *International Conference on Document Analysis and Recognition (ICDAR)*, 2026. Accepted for oral presentation. *Corresponding author. [Code]

[P2] Juena Noshin, Kawser Irom Rushee, Mohammad Rabiul Islam, Sifat Rahman Ahona, **Md Ashikur Rahman**. “UCAR: Uncertainty-Calibrated Adaptive Retrieval for Hallucination Reduction in Medical Vision-Language Models.” *International Conference on Computing Advancements (ICCA)*, 2026.

Preprints and Manuscripts Under Review

[M1] **Md Ashikur Rahman**, Md Arifur Rahman, Niamul Hassan Samin, Khandaker Rifah Tasnia, Sifat Rahman Ahona, Juena Ahmed Noshin. “Beyond Aggregate Risk: Role-Stratified Conformal Risk Control for LLM Tool Calls.” Manuscript under review, 2026.

[M2] **Md Ashikur Rahman**, Juena Ahmed Noshin, Niamul Hassan Samin, Abdullah Ibne Hanif Arean, Md Hasibul Amin, Md Arifur Rahman. “When Detector-Based Grounding Metrics Measure Vocabulary: A Cautionary Audit of Entity Claims in Video-QA Reasoning Traces.” Manuscript under review, 2026.

[M3] **Md Ashikur Rahman**, Md Arifur Rahman, Nusrat Jahan Trisna, Juena Ahmed Noshin. “Decoding Harm: Do Reasoning Models Resist Math-Encoded Jailbreaks?” Manuscript under review, 2026.

[M4] Niamul Hassan Samin, Abdullah Ibne Hanif Arean, Md Arifur Rahman, Md Hasibul Amin, Renu Akter Suity, Juena Ahmed Noshin, **Md Ashikur Rahman**. “Residual Stream Rebalancing: Training-Free Hallucination Mitigation in Vision-Language Models.” Manuscript under review, 2026.

Additional Publication

[A1] **Md Ashikur Rahman**, Md Arifur Rahman, Juena Ahmed Noshin. “Automated Detection of Diabetic Retinopathy Using Deep Residual Learning.” *International Journal of Computer Applications*, 2020.

Education

B.Sc. in Computer Science and Engineering

American International University-Bangladesh | 2011-2015

CGPA: 3.87/4.00; WES equivalent: 3.94/4.00

Magna Cum Laude; Top 3%; Merit Scholarship and Tuition Fee Waiver

Awards

• Champion, BASIS National ICT Awards 2020 (Retouched.ai); Finalist, APICTA 2021

Languages

• Bengali (Native); English (Professional working proficiency)

Skills

Multimodal and Vision-Language AI: Python, PyTorch, Vision-Language Models, Visual Grounding, Hallucination Evaluation, Multimodal Learning

Generative AI and LLMs: Diffusion Models, Large Language Models (including Llama and Qwen), Prompt Safety, Parameter-Efficient Fine-Tuning (LoRA, QLoRA)

Computer Vision: Object Detection, Segmentation, Pose Estimation, Image Quality Assurance

3D Vision: Structure from Motion, Multi-View Stereo, COLMAP, Open3D, Neural Radiance Fields (NeRF), 3D Gaussian Splatting

Production ML and MLOps: Model Serving, Deployment, Offline and Online Evaluation, A/B Testing, Dataset Design

Backend and Cloud: FastAPI, Docker, Nginx, Redis, Google Cloud Platform, Google Cloud Storage, SQL Server